

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — FULL STACK ACTIVE (STAGES 1-6)

Six-Stage AI-Native Design Life Cycle

1. Formulate & Explore

- Core Architecture Question
- Scoped Plan Contract
- Multi-Candidate DSE
- Search Frontier Freeze

2. Represent & Evaluate

- Machine-Readable Spec
- Executable Candidate Reps
- Independent Simulator Runs
- Metric Return Extraction

3. Interpret & Explain

- Empirical Observation Card
- Mechanism Attribution Check
- Failure Triage Diagnosis
- Evidence Package Freeze

4. Review & Decide

- Design-Loop Card Signoff
- Senior Architect Review

Verdict Gate:

ADVANCE / REJECT

Stage-Gated Provenance & Re-entry Loop